CHAPTER 1

Stochastic convergence

Asymptotic Statistics · MATH 598 · Mehdi Dagdoug ·

In this chapter, we define four important modes of stochastic convergence: almost sure, in probability, in L^p and in distribution. Some of these modes of convergence are stronger than others; we will discuss these relationships. Although some of these modes of convergence are weaker, they sometimes can be combined with other arguments to imply their stronger counterparts; we call these implications partial converses. Often, the convergence properties of various difficult statistics can be deduced by establishing the convergence of several easier statistics, before combining them. Tools like the continuous mapping theorem and Slutsky will be derived. We then conclude by introducing popular (and convenient) stochastic order notations and their algebra.

Notations

Throughout the course, we work on a probability space $(\Omega, \mathcal{A}, \mathbb{P})$. A random vector X is a measurable map $X : \Omega \to \mathbb{R}^d$ for $d \geq 1$; we say that it is a random variable if $d = 1$. To be most useful, our definitions will often be provided for random vectors and we will see random variables as particular cases. When nothing is lost, we will sometimes work with random variables directly. The dimension should be clear from the context.

The random vector X induces a new law[1] $\mathbb{P}_X := \mathbb{P} \circ X^{-1}$ on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$, that is,

$$\mathbb{P}\left[X \in A\right] = \mathbb{P}[\{\omega \in \Omega : X(\omega) \in A\}] = \mathbb{P}\left[X^{-1}(A)\right] = \mathbb{P}_X(A), \qquad A \in \mathcal{B}(\mathbb{R}^d),$$

where $\mathcal{B}(\mathbb{R}^d)$ denotes the Borel sigma-algebra[2] on $\mathbb{R}^d$.

We will denote by $\|x\| := (\sum_{j=1}^d x_j^2)^{1/2}$, the Euclidean norm of a vector $x \in \mathbb{R}^d$. The distribution function of a random vector X is denoted

$$F_X : t \mapsto \mathbb{P}[X \leq t] = \mathbb{P}[X_{(1)} \leq t_{(1)}, ..., X_{(d)} \leq t_{(d)}], \qquad t \in \mathbb{R}^d.$$

The law of X is denoted by $\mathcal{L}(X)$. For example, if $X \sim \mathcal{N}(0,1)$, then $\mathcal{L}(X) = \mathcal{N}(0,1)$. The expectation of X, defined when $\mathbb{E}[\|X\|] < \infty$, is given by

$$\mathbb{E}[X] = \int_\Omega X(\omega)\mathbb{P}(\mathrm{d}\omega) = \int_{\mathbb{R}^d} x\mathbb{P}_X(\mathrm{d}x).$$

[1] In this text, the word "law" is synonymous with "distribution".

[2] A Borel sigma-algebra on a (topological) space E is the sigma-algebra generated by the open sets of E.

1.1 The main modes of stochastic convergence

The starting point of our discussion on convergence of random variables is to notice that random variables are functions, in the world of probability. We typically will say that a sequence of vectors $(x_n)_{n\in\mathbb{N}}$ converges to a point x if for all $\epsilon > 0$, there exists an integer $n_0 := n_0(\epsilon) \in \mathbb{N}$ such that, for all $n \geq n_0$, $\|x_n - x\| \leq \epsilon$. That is, for any given threshold (ϵ), there exists a moment (n_0) after which, the sequence $(x_n)_{n\in\mathbb{N}}$ will stay close to x (within an ϵ neighborhood).

However, *random variables are functions*. Two types of convergence of sequences of functions are most common: pointwise and uniform convergence. Let $(f_n)_{n\in\mathbb{N}}$ be a sequence of functions $f_n : A \to \mathbb{R}^d$ and $f : A \to \mathbb{R}^d$ be another function. We say that f_n converges pointwise to f on A if, for every fixed $x \in A$, the sequence $f_n(x)$ converges to $f(x)$ (note, this is now a sequence of vectors). Uniform convergence of f_n to f, on the other hand, means that $\sup_{x\in A}\|f_n(x) - f(x)\| \to 0$ as $n \to \infty$. Uniform convergence is stronger than pointwise convergence (it implies it). However, although these notions of convergence of sequences of functions are useful, *random variables are functions in a world of probability*. We will therefore combine convergence of functions with probability statements.

Although more types of convergence of random variables exist, in statistics, only four types will be useful in these notes: almost sure convergence, convergence in probability, in L^p, and in law. These will be the focus of this section.

1.1.1 Almost sure convergence

Almost sure convergence is perhaps the most natural way to define convergence of a sequence of random variables, in that it is a natural extension of the deterministic setting. Almost sure convergence is also called convergence with probability one[3].

Definition 1.1 (Almost sure convergence).
A sequence of random vectors $(X_n)_{n\in\mathbb{N}}$ ***converges almost surely*** to a random vector X, denoted $X_n \xrightarrow[n\to\infty]{a.s.} X$, if

$$\mathbb{P}\left[\omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)\right] = 1.$$

There are various ways to rewrite the definition of almost sure convergence. The next lemma gives useful equivalences. Before proceeding, we introduce the following notions: infinitely often, finitely many, and limits of sets. These will be needed to further describe almost sure convergence.

Lemma 1.2 (Almost sure convergence: equivalences).
The following statements are equivalent.

(i) $X_n \xrightarrow[n\to\infty]{a.s.} X$.

(ii) There exists an event $E \in \mathcal{A}$, such that $\mathbb{P}(E) = 1$ and X_n converges to X pointwise on E.

[3] In general, "with probability one" (wp1), "almost surely" (a.s.), or even "almost everywhere" (a.e.), all have the same meaning. Specifically, a statement S is said to hold a.s. on $(\Omega, \mathcal{A}, \mathbb{P})$ if there exists an event $E \in \mathcal{A}$ such that S holds for all $\omega \in E$ and $\mathbb{P}(E) = 1$.

(iii) For all $\epsilon > 0$, $\mathbb{P}\left[\|X_n - X\| > \epsilon \text{ i.o.}\right] = 0$.

(iv) For all $\epsilon > 0$, $\lim_{n\to\infty} \mathbb{P}\left[\sup_{k\geq n}\|X_k - X\| > \epsilon\right] = 0$.

Proof. Proof of (i) $\Leftrightarrow$ (ii). We let $E := \{\omega \in \Omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)\}$. By definition $X_n \xrightarrow[n\to\infty]{a.s.} X$ if and only if $\mathbb{P}(E) = 1$. Moreover, on E, we have $\lim_{n\to\infty} X_n(\omega) = X(\omega)$, which is the definition of pointwise convergence of X_n to X on E.

Proof of (i) $\Leftrightarrow$ (iii). Let $\epsilon > 0$ and define $A_\epsilon := \{\|X_n - X\| > \epsilon, \text{ i.o.}\}$. We will use the following fact.

Fact. A sequence $(u_n)_{n\in\mathbb{N}}$ of nonnegative numbers satisfies $\lim_{n\to\infty} u_n = 0$ iff, for every $\epsilon > 0$, $u_n > \epsilon$ only finitely many times.

Leveraging this fact, we see that

$$\{\|X_n - X\| \to 0\} = \bigcap_{\epsilon>0} A_\epsilon^c.$$

Technique. Observe that

$$\bigcap_{\epsilon>0} A_\epsilon^c = \bigcap_{m\geq 1} A_{1/m}^c.$$

Replacing the uncountable intersection over all $\epsilon > 0$ by a countable intersection is interesting because countable intersections are measurable, uncountable need not be. Here, therefore, convergence to 0 can be tested on the countable set $\{1/m : m \in \mathbb{N}^*\}$.

Finally, to conclude, we leverage the following fact.

Fact. If a countable intersection has probability one, each event still has probability one.

Therefore, we have

$$\begin{aligned}
\mathbb{P}\left[\|X_n - X\| \to 0\right] = 1 \quad &\Leftrightarrow \quad \mathbb{P}\left[\bigcap_{m\geq 1} A_{1/m}^c\right] = 1 \\
&\Leftrightarrow \quad \mathbb{P}\left[A_{1/m}\right] = 0, \quad \text{for all } m \geq 1, \\
&\Leftrightarrow \quad \mathbb{P}\left[A_\epsilon\right] = 0, \quad \text{for all } \epsilon > 0.
\end{aligned}$$

Proof of (iii) $\Leftrightarrow$ (iv). Let $\epsilon > 0$ and $B_n := \{\sup_{k\geq n}\|X_k - X\| > \epsilon\}$, and note that

$$B_n = \bigcup_{k\geq n} \{\|X_k - X\| > \epsilon\}.$$

The events B_n are decreasing, so $B_n \downarrow \cap_{n\geq 1} B_n$. Hence, by continuity from above,

$$\lim_{n\to\infty} \mathbb{P}[B_n] = \mathbb{P}\left[\bigcap_{n\geq 1} B_n\right] = \mathbb{P}\left[\bigcap_{n\geq 1}\bigcup_{k\geq n} \{\|X_k - X\| > \epsilon\}\right] = \mathbb{P}\left[\|X_n - X\| > \epsilon \text{ i.o.}\right].$$

■

Intuitively, the idea of almost sure convergence is that, for almost all realized outcomes ω, the numerical sequence $X_1(\omega), X_2(\omega), X_3(\omega), ...$ converges to $X(\omega)$; this is therefore a *pathwise* notion of convergence. So, with probability one, X_n is eventually close to X, and stays close forever.

Exercise 1.1 (★★).
Let $U_1, U_2, \ldots$ be independent random variables with law $\mathcal{U}([0,1])$ and let $M_n := \max_{1\leq k\leq n} U_k$.

(a) Show that, for almost every $\omega \in \Omega$, the sequence $(M_n(\omega))_{n\in\mathbb{N}}$ is nondecreasing and bounded, hence converges to some $M_\infty(\omega) \in [0,1]$.

(b) For $\epsilon \in (0,1)$, compute $\mathbb{P}[M_n \leq 1-\epsilon]$.

(c) Show that $M_n \xrightarrow[n\to\infty]{a.s.} 1$. *Hint:* write $\{M_\infty \leq 1-\epsilon\}$ as a decreasing limit of events, then use $\{M_\infty < 1\} = \bigcup_{m\geq 1}\{M_\infty \leq 1 - 1/m\}$.

1.1.2 Convergence in probability

We now turn to another, very important, notion of convergence of random variables.

Definition 1.3 (Convergence in probability).
A sequence of random vectors $(X_n)_{n\in\mathbb{N}}$ ***converges in probability*** to a random vector X, denoted $X_n \xrightarrow[n\to\infty]{\mathbb{P}} X$, if for all $\epsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}\left[\|X_n - X\| > \epsilon\right] = 0.$$

Intuitively, convergence in probability means that, for large n, X_n is very likely to be close to X. It is not a pathwise statement, but says that the probability of a large error is small, as n gets large.

Some examples. As a simple example, consider the case where $Z \sim \mathcal{N}(0,1)$ and $X_n := X + Z/n$. Then, the claim is that $X_n \xrightarrow[n\to\infty]{\mathbb{P}} X$. To see this, let $\epsilon > 0$. Then,

$$\mathbb{P}\left[|X_n - X| > \epsilon\right] = \mathbb{P}\left[|Z| > n\epsilon\right].$$

Now, as $n \to \infty$, we have $n\epsilon \to \infty$ so that the probability of $|Z|$ being greater than $n\epsilon$ decreases to 0. To see this more formally, note that the sequence of events $(\{|Z| > n\epsilon\})_{n\in\mathbb{N}}$ is decreasing to $\cap_{n\in\mathbb{N}^*}\{|Z| > n\epsilon\} = \varnothing$, and thus, by continuity from above,

$$\lim_{n\to\infty} \mathbb{P}\left[|Z| > n\epsilon\right] = \mathbb{P}\left[\lim_{n\to\infty}\{|Z| > n\epsilon\}\right] = \mathbb{P}\left[\varnothing\right] = 0.$$

This shows therefore that convergence in probability holds. We plot the densities of X_n around X for various choices of n with two different threshold boxes of ϵ in Figures 1.1 and 1.2.

It is important to note that it does not forbid rare extreme values. For example, consider the sequence

$$X_n = \begin{cases} n, & \text{with probability } \dfrac{1}{n}, \\ 0, & \text{with probability } 1 - \dfrac{1}{n}. \end{cases}$$

Then, for $\epsilon > 0$ and sufficiently large n, we have

$$\mathbb{P}\left[|X_n - 0| > \epsilon\right] = \mathbb{P}\left[X_n = n\right] = n^{-1},$$

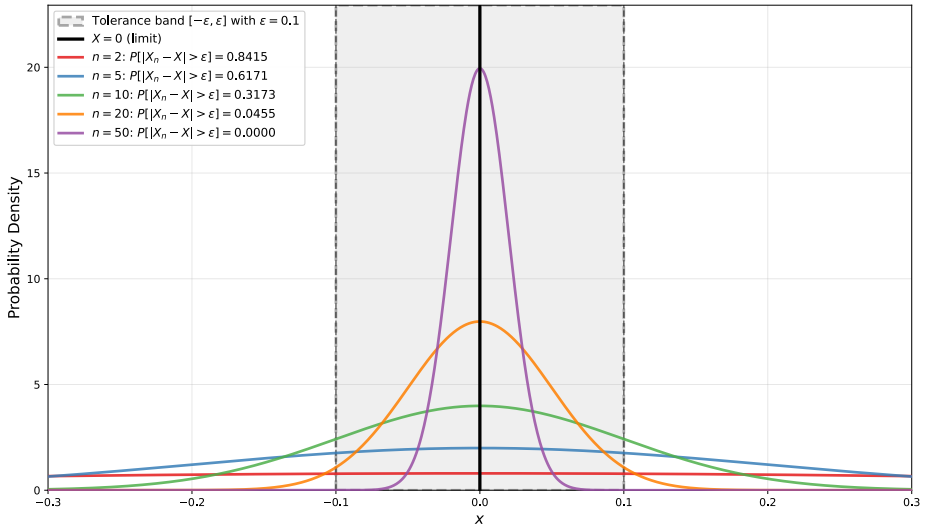


Figure 1.1. Varying densities of X_n around $X = 0$ as n varies with $\epsilon = 0.1$.

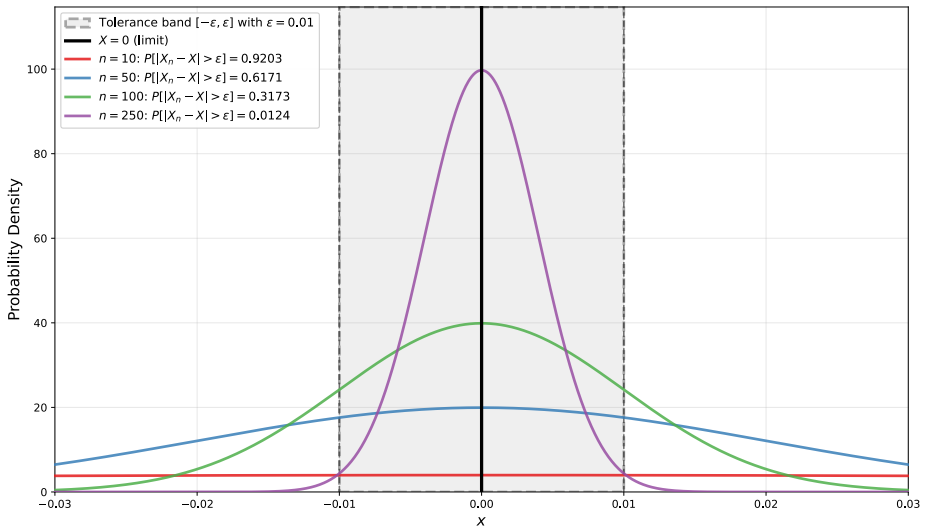


Figure 1.2. Varying densities of X_n around $X = 0$ as n varies with $\epsilon = 0.01$.

which shows that $X_n \xrightarrow[n\to\infty]{\mathbb{P}} 0$. However, it is also true that, with vanishingly small probability, X_n can take larger and larger values as $n \to \infty$. As we will see later on, this may or may not prevent other types of convergence from happening.

1.1.3 Convergence in distribution

We now introduce a weaker type of convergence, with many important uses in statistics: convergence in distribution. It is also called ***convergence in law*** or ***weak convergence***. To that aim, let us introduce the following notations. Recall that F_X denotes the cumulative distribution function of a random vector X. For a function f, we write $C(f) := \{t \in \mathrm{dom}(f) : f \text{ is continuous at } t\}$ the set of its continuity points. Moreover, we write $C_b(\mathbb{R}^d)$ to denote the set of bounded continuous functions on $\mathbb{R}^d$.

Definition 1.4 (Convergence in distribution).
A sequence of random vectors $(X_n)_{n\in\mathbb{N}}$ ***converges in distribution*** to a random vector X, denoted $X_n \xrightarrow[n\to\infty]{\mathcal{L}} X$, if

$$\lim_{n\to\infty} F_{X_n}(t) = F_X(t), \quad \text{for all } t \in C(F_X),$$

where $C(F_X) := \{t \in \mathrm{dom}(F_X) : F_X \text{ is continuous at } t\}$.

Exercise 1.2 (★★).

(a) Suppose that $X_n \xrightarrow[n\to\infty]{\mathbb{P}} X$ and $X_n \xrightarrow[n\to\infty]{\mathbb{P}} Y$. Show that $\mathbb{P}[X = Y] = 1$.

(b) Suppose that $X_n \xrightarrow[n\to\infty]{a.s.} X$ and $X_n \xrightarrow[n\to\infty]{a.s.} Y$. Show that $\mathbb{P}[X = Y] = 1$.

(c) Let $X \sim \mathcal{N}(0,1)$ and $X_n := X$ for all n. Show that $X_n \xrightarrow[n\to\infty]{\mathcal{L}} X$ and $X_n \xrightarrow[n\to\infty]{\mathcal{L}} -X$, although $\mathbb{P}[X = -X] = 0$. What is the correct uniqueness statement for convergence in distribution?

Convergence in distribution is somewhat separate from the previous types of convergence, in that it does not require X_n to get close to X in the traditional sense. First, we can note that the definition does not require X_n and X to be defined on the same probability space. Even if they were, it also does not require X_n to be close to X, but that *the distribution of X_n is getting closer to that of X*. It is therefore really a statement about the law of X_n resembling more and more that of X.

Specifically, convergence in distribution requires pointwise convergence of the sequence of functions $(F_{X_n})_{n\in\mathbb{N}}$ to F_X on the set $C(F_X)$. Therefore, pointwise convergence is required only at continuity points of the target CDF F_X.

Some examples. First, let $X_n = -X$ where $X \sim \mathcal{N}(0,1)$. Then, although surprising, $X_n \xrightarrow[n\to\infty]{\mathcal{L}} X$. Indeed, because X is standard Gaussian, its distribution is symmetric[4], so that $\mathcal{L}(X_n) =$

[4] A random variable X has a symmetric distribution (around 0) if $X \overset{d}{=} -X$, that is, the law of X is the same as that of $-X$, i.e., $\mathcal{L}(X) = \mathcal{L}(-X)$. A normal distribution is symmetric around μ.

$\mathcal{L}(-X) = \mathcal{L}(X)$, and thus, for any $n \in \mathbb{N}$, for any $t \in \mathbb{R}$, $F_{X_n}(t) = F_X(t)$. However, when $X(\omega) = x$, $X_n(\omega) = -x$, so X_n can be quite far from X, and this for any n.

Another simple example is the following. Let $X_n = X + n^{-1}$, for some random variable X. Then, let $t \in C(F_X)$ and write

$$F_{X_n}(t) = \mathbb{P}\left[X_n \leq t\right] = \mathbb{P}\left[X \leq t - n^{-1}\right] = F_X(t - n^{-1}).$$

Then, $t - n^{-1} \to t$ as $n \to \infty$ and since t is a continuity point of F_X, we get $\lim_{n\to\infty} F_X(t - n^{-1}) = F_X(t)$, which shows convergence in distribution. We plot the CDFs of X_n and X with varying sample sizes in Figure 1.3.

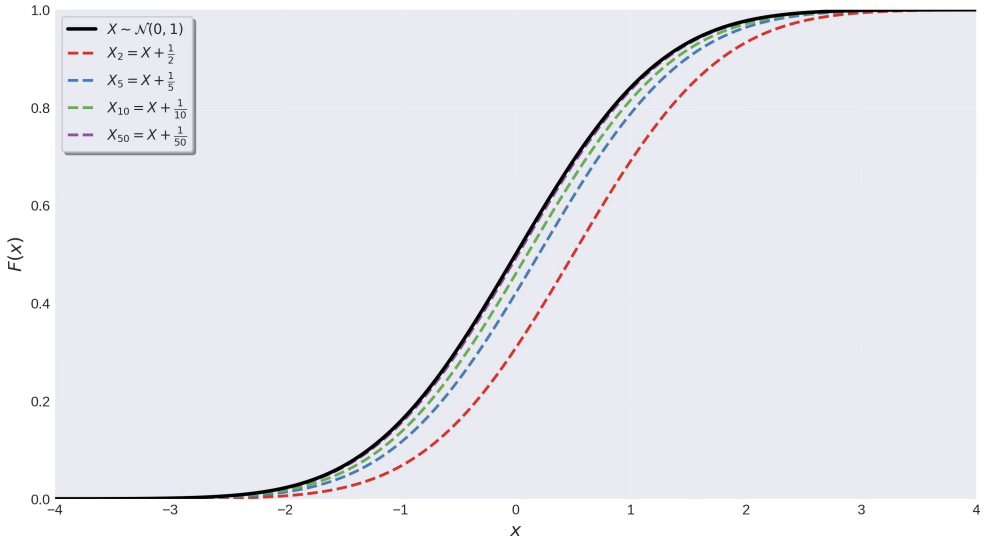


Figure 1.3. Varying CDFs of X_n and X as n varies.

Equivalent reformulations. Although we defined convergence in distribution using CDFs, there are various other possibilities to define it. A well-known lemma, the ***Portmanteau Lemma***, gives several useful equivalences to convergence in distribution.

Before proceeding, let us recall briefly some topological notions. For $x \in \mathbb{R}^d$ and $r > 0$, we denote by $\mathcal{B}(x, r) := \{y \in \mathbb{R}^d : \|x - y\| < r\}$ the *open ball* centered at x with radius r. We say that a set $A \subseteq \mathbb{R}^d$ is *open* if, for every $x \in A$, there exists $r > 0$ such that $\mathcal{B}(x, r) \subset A$. The *interior* of a set A is

$$A^\circ := \{x \in A : \text{there exists } r > 0 \text{ such that } B(x, r) \subseteq A\}.$$

Thus, A° is the largest open set contained in A. The set A is *closed* if its complement A^c is open. The *closure* of A is the set

$$\overline{A} := \left\{x \in \mathbb{R}^d : B(x, r) \cap A \neq \varnothing \text{ for every } r > 0\right\}.$$

Thus, $\overline{A}$ is the smallest closed set containing A. Informally, it is meant to contain the points of A, and those that can be approached arbitrarily closely by points of A. Equivalently, a point $x \in \overline{A}$ if and only if there exists a sequence $(x_n)_{n\in\mathbb{N}} \subset A$ such that $\lim_{n\to\infty} x_n = x$. The *boundary* of A is

$$\partial A = \overline{A} \setminus A^\circ.$$

For example, $(a, b)^\circ = (a, b)$, $\overline{(a, b)} = [a, b]$, and $\partial(a, b) = \{a, b\}$.

Lemma 1.5 (Portmanteau)**.**
The following statements are equivalent.

(i) $X_n \xrightarrow[n\to\infty]{\mathcal{L}} X$.

(ii) For every $f \in C_b(\mathbb{R}^d)$,
$$\lim_{n\to\infty} \mathbb{E}\left[f(X_n)\right] = \mathbb{E}\left[f(X)\right].$$

(iii) For every closed set $F \subseteq \mathbb{R}^d$,
$$\limsup_{n\to\infty} \mathbb{P}\left[X_n \in F\right] \leq \mathbb{P}\left[X \in F\right].$$

(iv) For every open set $G \subseteq \mathbb{R}^d$,
$$\liminf_{n\to\infty} \mathbb{P}\left[X_n \in G\right] \geq \mathbb{P}\left[X \in G\right].$$

(v) For every Borel set $A \subseteq \mathbb{R}^d$ such that $\mathbb{P}\left[X \in \partial A\right] = 0$, we have
$$\lim_{n\to\infty} \mathbb{P}\left[X_n \in A\right] = \mathbb{P}\left[X \in A\right].$$

Proof. See van der Vaart (1998), page 6. ■

References

A. W. van der Vaart. *Asymptotic Statistics*. Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge University Press, 1998.